

Standard Error

Conceptually, **Standard Error** is the inferential statistics counterpart to standard deviation.

Inferential Statistics

Standard Error is a measure of the variability of a sample statistic in our population.

Descriptive Statistics

Standard Deviation is a measure of the variability of an individual measurement in our sample.

To calculate standard error, we can use a technique called **bootstrapping**:

Assuming that our sample is representative of our population:

1. Draw 100-300 *resamples* from our sample.
 - With replacement
 - Same size as sample
2. Calculate the **sample statistic** (e.g., mean) of each resample.
3. Double check that the distribution of the sample statistics is normally distributed.
 - Thanks to **Central Limit Theorem**, this is usually true.
4. The standard deviation of the sample statistics is our population standard error.

Demo

Calculate Standard Error.



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